Sifatul Anindho

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Research Interests

Multimodal foundation models, vision-language reasoning, model abstention and selective prediction, post-training, reinforcement learning, intervention timing, and human-centered AI.

Education

Colorado State University — Fort Collins, CO
Ph.D. in Computer Science, GPA: 4.00/4.00 — *Expected May 2029* Advisor: Nikhil Krishnaswamy

Colorado State University — Fort Collins, CO
M.S. in Computer Science, GPA: 4.00/4.00 — *July 2026* Advisor: Nathaniel Blanchard

Michigan State University — East Lansing, MI
B.S. in Computer Science, GPA: 3.81/4.00 — *May 2024*

Research and Engineering Experience

SIGNAL Lab, Colorado State University — Fort Collins, CO
Graduate Research Assistant, Advisor: Nikhil Krishnaswamy — *Aug. 2026–Present*

- Investigate evidence-aware answering, abstention, and intervention timing for vision-language models in multimodal task guidance.
- Develop evaluations for unsupported answering and insufficient visual evidence in lower-limb ultrasound guidance.
- Lead continued development and integration of the lower-limb DVT use case within the ARPA-H PARADIGM program.

IMPACT Lab, Colorado State University — Fort Collins, CO
Graduate Research Assistant, Advisor: Nathaniel Blanchard — *Aug. 2024–July 2026*

- Designed participant-grounded retrospective cued-recall methods for multimodal cognitive-affective state annotation in collaborative and clinical tasks.
- Built multimodal datasets aligning video, audio, transcripts, facial behavior, and self-reported internal states.
- Evaluated facial, acoustic, linguistic, and temporal models using group-disjoint validation and analyzed ordered state transitions.
- Contributed to the DARPA FACT and ARPA-H PARADIGM research programs.

HLR Lab, Michigan State University — East Lansing, MI
Undergraduate Research Assistant, Advisor: Parisa Kordjamshidi — *May 2024–July 2024*

- Developed an LLM-supported natural-language interface for constructing complex neuro-symbolic models with DomiKnowS.

Whirlpool — Benton Harbor, MI
Senior Design Intern — *Jan. 2024–May 2024*

- Developed software for a human-machine interface supporting personalized interaction with Whirlpool smart ovens.

Liu Lab, Michigan State University — East Lansing, MI
Undergraduate Research Assistant, Advisor: Kevin Liu — *May 2023–Aug. 2023*

- Investigated the empirical performance of phylogenetic-estimation algorithms using tree-space analyses.

Selected Research and Engineering Projects

VIGIL: Real-Time Multimodal Guidance for Lower-Limb DVT Ultrasound — *May 2025–Present*
ARPA-H PARADIGM

- Implemented major components of a real-time ROS2 guidance platform supporting live ultrasound sessions and reproducible offline evaluation from recorded ROS bags.
- Integrated VQA, probe-state detection, task-state tracking, and vein-detection models into an end-to-end multimodal guidance pipeline.
- Wired outputs from multiple ROS nodes into context-rich VQA prompts for procedural guidance generation.
- Profiled latency across ultrasound processing, model inference, VQA generation, and interface rendering to diagnose system bottlenecks.
- Led recurring integration and testing across independently developed components, supporting site visits, research demonstrations, and annual external evaluation by MIT Lincoln Laboratory.
- Identified hallucination, overfitting, and unsupported-answering failures during integration, motivating current research on training VLMs to recognize insufficient visual evidence.
- Collaborated with researchers and engineers across Colorado State University, University of Michigan, University of Pennsylvania, and Cornell Tech.

FACT: Friction for Accountability in Conversational Transactions — *Aug. 2024–Aug. 2025*
DARPA FACT

- Designed a retrospective cued-recall framework for synchronizing participant-reported cognitive-affective trajectories with multimodal interaction data.
- Developed temporal-alignment and fuzzy sequence-labeling procedures for uncertain internal-state boundaries.
- Built facial-behavior and multimodal feature-extraction pipelines for studying AI-assistant intervention effectiveness.

Peer-Reviewed Publications

- **S. Anindho**, et al. "Ordered Network Analysis of Epistemic Emotions during Collaborative Problem Solving." *International Conference on Artificial Intelligence in Education (AIED)*, 2026.
- V. Venkatesha, et al. "Using LLMs to Annotate Pedagogical Moves: You Know What I Mean?" *International Conference on Artificial Intelligence in Education (AIED)*, 2026.
- **S. Anindho**, et al. "Evaluating Web-Trained Facial Expression Recognition in Collaborative Problem Solving." *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2026.
- E. Seefried, et al. "CSU101: An Educational Dataset for Introductory Computer Vision." *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2026.
- V. Venkatesha, et al. "Detecting Self-Reported Cognitive-Affective States in Collaborative Problem Solving from Prosodic and Linguistic Features." *Educational Data Mining (EDM), Poster/Demo Track*, 2026.
- **S. Anindho** and N. Blanchard. "Identifying and Modeling Epistemic Emotions in Collaborative Problem Solving." *Educational Data Mining (EDM), Doctoral Consortium*, 2026.
- **S. Anindho**, et al. "A Methodological Framework for Capturing Cognitive-Affective States in Collaborative Learning." *Multimodal, Multiparty Learning Analytics Workshop at Educational Data Mining (EDM)*, 2025.
- **S. Anindho**, et al. "An Exploration of Internal States in Collaborative Problem Solving." *International Conference on Human-Computer Interaction (HCII)*, pp. 135–150, 2025.

Open-Source Software

TRACE: Transparency, Reflection, and Accountability in Collaborative Exchanges *June 2025–Mar. 2026*
Used by DARPA FACT and NSF iSAT
- Contributed to a reusable platform for constructing and running complex multimodal human-AI interaction demonstrations.

Lab Operations Scheduler *Sept. 2025–Dec. 2025*
Used by the CSU Department of Computer Science
- Built an integer-programming system that assigned 50 lab operators to time slots under availability and preference constraints.

Multi-Label Video Annotation Tool *Oct. 2025*
Used by ARPA-H PARADIGM
- Built a desktop application for efficient multi-label annotation of research videos.

Retrospective Cued-Recall Annotation Platform *Feb. 2025*
Used by DARPA FACT
- Built a web-based platform supporting self-caught and probe-caught retrospective reporting synchronized with session video.

Selected Research Presentations and Demonstrations

ARPA-H VIGIL Site Visit, 2025; ARPA-H PARADIGM PI Meeting, 2025; DARPA FACT PI Meeting, 2025; International Conference on Human-Computer Interaction, 2025; Mid-Michigan Symposium for Undergraduate Research Experiences, 2024.

Teaching Experience

Colorado State University Fort Collins, CO
Graduate Teaching Assistant *Aug. 2024–Present*
Courses: Introduction to Machine Learning; Machine Learning Foundations and Practice; Software Development; Computational Thinking with Java.

Michigan State University East Lansing, MI
Undergraduate Teaching Assistant *Jan. 2022–May 2024*
Course: Algorithmic Thinking and Programming; five semesters.

Academic Service

Reviewer: ACL 2025, ACL 2026, CVPRW 2026 | Graduate Program Committee Representative, CSU, 2026 | Judge, CSU CURC, 2026

Honors and Awards

Outstanding Cambridge Learner Award, University of Cambridge, 2018 — Highest score in the world in AS Level Mathematics.
Global Spartan Scholarship, Michigan State University, 2021 — Awarded to top-achieving international undergraduates.

Technical Skills

Languages: Python, C++, Java, JavaScript/TypeScript, SQL, R, Bash, MATLAB
Machine Learning and AI: PyTorch, TensorFlow, JAX, Hugging Face Transformers, TRL, scikit-learn, PyTorch Lightning, XGBoost, OpenCV, pandas, NumPy, SciPy
Methods: LLMs, VLMs, multimodal learning, reinforcement learning, preference optimization, fine-tuning, representation learning, uncertainty estimation, selective prediction, experimental design, statistical modeling, time-series analysis
Systems and MLOps: ROS2, Docker, Kubernetes, Linux, Git, CI/CD, MLflow, Weights & Biases, FastAPI, Flask, real-time inference, profiling, performance optimization
Cloud and Data: AWS, GCP, Azure, SageMaker, Spark, PostgreSQL, MySQL, MongoDB